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Business Intelligence

Week 8-9 – GEITE01X – Living in the IT Era

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Learning Outcomes

At the end of the lesson the student should be able to:

1. Define what is Business Intelligence (BI) and Analytics
2. Enumerate the different BI/Analytics systems and tools
3. Know the BI evolution and trend

What is Business Intelligence?

Business Intelligence is a set of methods, processes, architectures, applications, and technologies that gather and transform raw data into meaningful and useful information used to enable more effective strategic, tactical, and operational insights and decision-making.

Data

Different types of data:

- Numeric vs. textual
- Structured vs. unstructured
- Standard format vs. proprietary format
- Internal vs. external data, system stored vs. file based data
- Raw fact data vs. simulated/forecast/estimated data
- Simple fact data vs. calculated metrics data

Common Data Problems

Common data problems:

- Structured, unstructured, semi-structured
- Information overloading
- Big data
- Data everywhere
- Difficulty of access
- Lack of data

Decision Making

- Decisions can be made based on
  - Facts, or data
  - Simulation (models)
  - Intuition, perception, sense
  - Group negotiation

Decision Making

- Traditionally BI has been also understood as Decision Support System (DSS) – known as data driven DSS (data directly contributes to decision without intensive and advanced analytical techniques).

Decision Making

- Problems in decision making
  - A gap between data and knowledge (useful information leading to a decision).
  - Management/operation by intuition
  - Lack of effective feedback and alignment systems, no improvement cycles
  - Need good analytical processing and models

Decision Making

- Evolving analytical needs in decision support
  - Real-time, most recent data
  - Business user driven, agile, instant
  - Exploratory and interactive

Evolution of Business Intelligence

1980s	Executive information systems (EIS), decision support systems (DSS)
1990s	Data warehousing (DW), business intelligence (BI)
2000s	Dashboards and scorecards, performance management
2010+??	Analytics, big data, data science, augmented BI, …

Analytics

- Analytics has emerged as a catch-all term for a variety of different business intelligence (BI) and application-related initiatives. … Whatever the use cases, "analytics" has moved deeper into the business vernacular.
- Analytics refers to a more systematical, automated, and flexible process of data analysis for revealing insights and decision support in more extensive application areas (beyond organizational contexts), e.g. sports, disease, network traffic, etc.

Depending on perspectives, Analytics
- is part of BI
- includes BI
- goes beyond (the traditional) BI
- = (the new) BI

Analytics or BI

- We tend to call analytics rather than BI in the following scenarios. But their processes and technologies are very similar.
- Non-business activities such as
  - Learning analytics
  - Talent analytics
  - Web analytics
  - Sports analytics
- Non-organizational contexts; mainly used by individuals or groups for public communication.

BI and other related terms

- Big data

– "Big Data is not a system; it is simply a way to say that you have a lot of data.

– Big data covers non-structure and various data formats including text, blob, multimedia, etc.

BI and other related terms

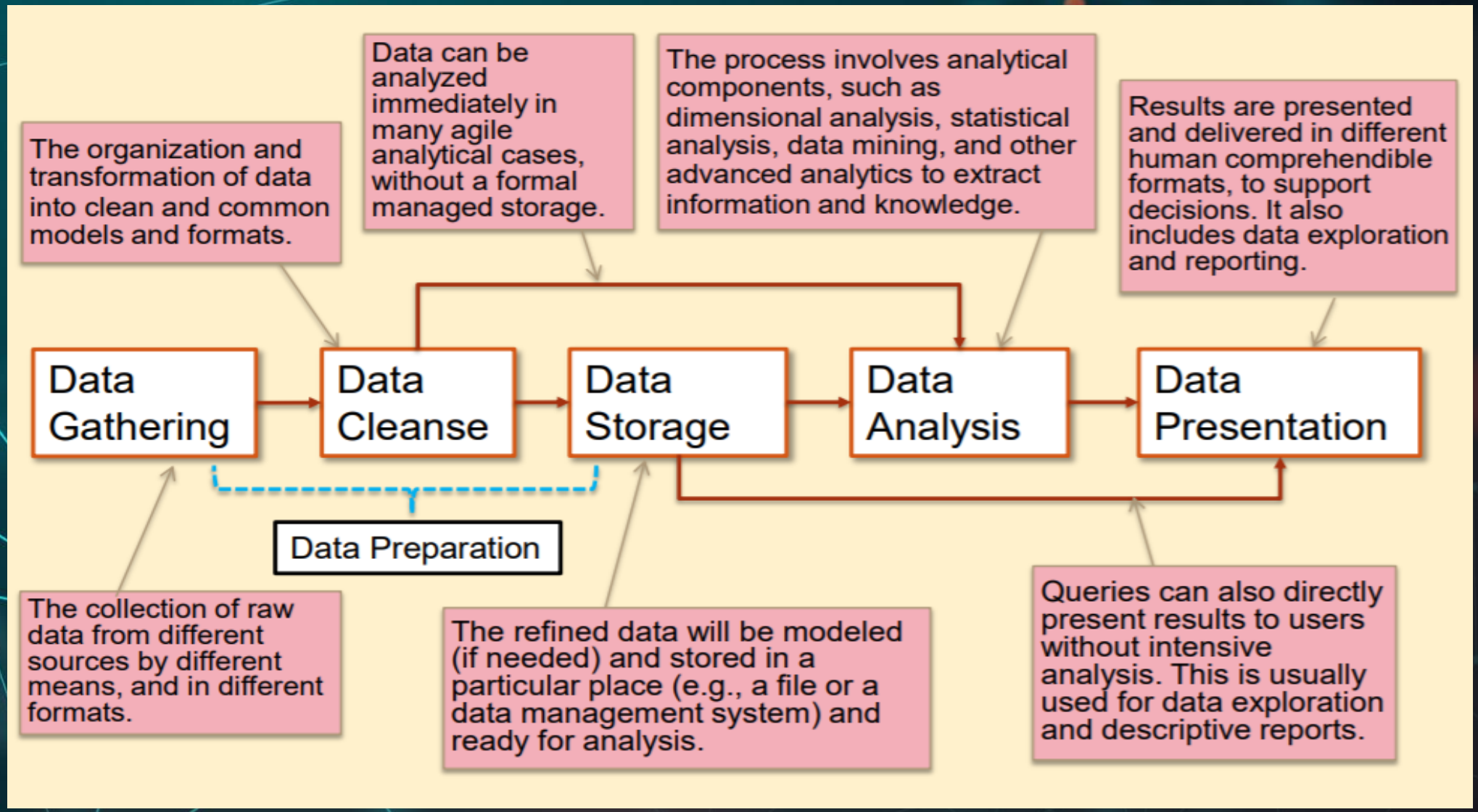
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- Data science

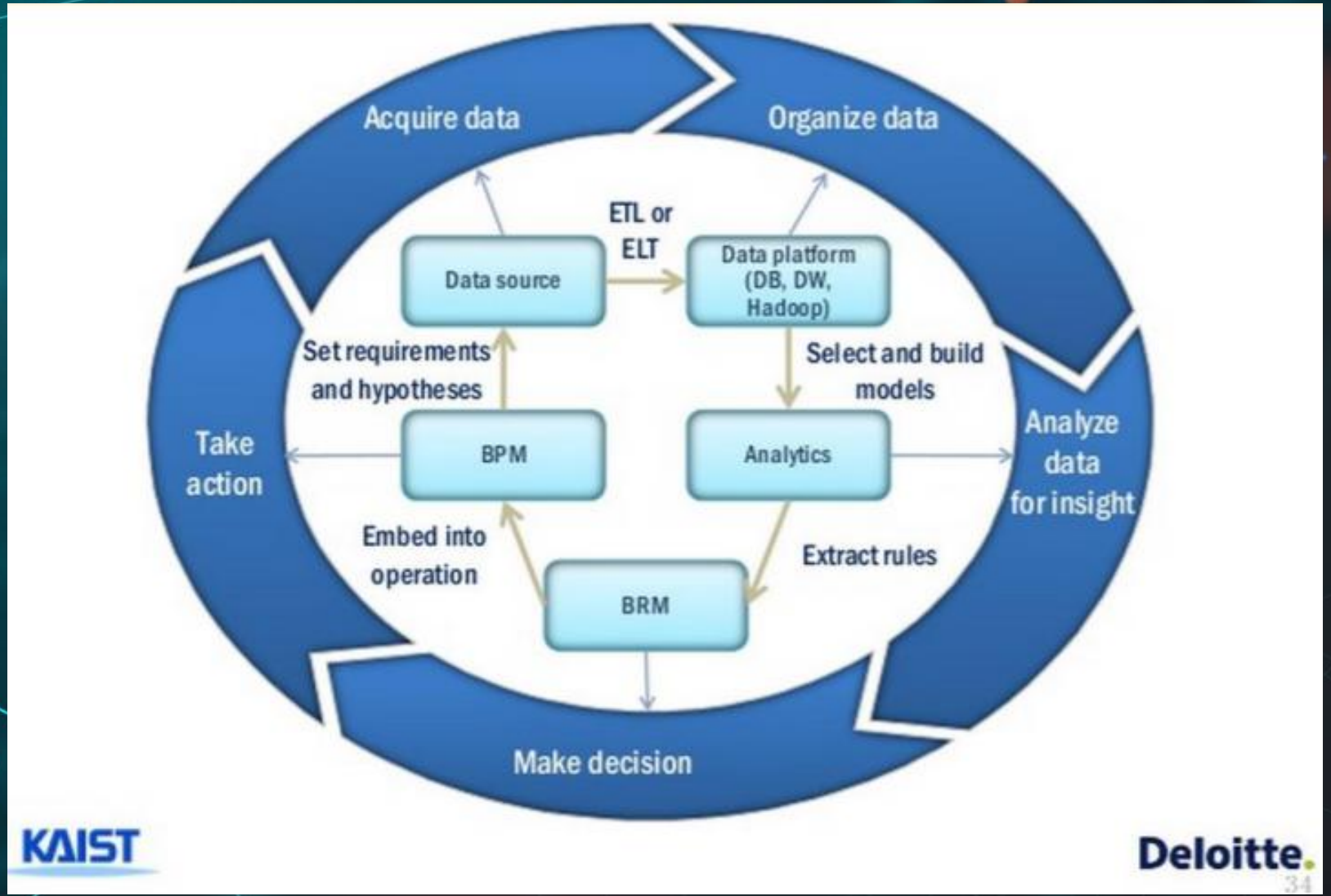
– An interdisciplinary field about processes and systems to extract knowledge or insights from data in various forms
– Focus on advanced analytics and presentation models and methods
– Using autonomous or semi-autonomous techniques and tools, typically beyond traditional BI to discover deeper insights, make predictions, or generate recommendation.
– A good data scientist = data hacker + programmer+ analyst + coach + story teller + artist
– “In some ways, data science is an evolution of BI.”

BI/Analytics: A General Process

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BI/Analytics: Decision Process



BI Systems and Platforms

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- A BI system is a computer information system that implements (part or whole) BI capabilities and processes
- The values of BI Systems
  - Provide an integrated data (analytical) processing platform
  - Enable easy and fast access of data and information at all levels (raw data, analysis results, metrics, etc.)
  - Streamline a controlled and managed process of data driven decision making
- Enterprise level vs. personal level
  - An enterprise level BI system emphasizes more on control and performance.
  - While a more user-oriented analytics platform enables nontechnical users to autonomously execute full-spectrum analytic workflows from data access and preparation to interactive analysis and the collaborative sharing of insights.

Critical Capabilities of a BI and Analytics Platform

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- Infrastructure
  - BI Platform Administration
  - Cloud BI
  - Data Source Connectivity
- Data Management
  - Governance and Metadata Management
  - Self Contained ETL and Data Storage
  - Self-Service Data Preparation
  - Scalability and Data Model Complexity

Critical Capabilities of a BI and Analytics Platform

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- Analysis and Content Creation
  - Advanced Analytics
  - Analytic Dashboards
  - Interactive Visual Exploration
  - Augmented Data Discovery
  - Mobile Exploration and Authoring
- Sharing and Findings
  - Embedding Analytic Content
  - Publish and Collaborate Analytic Content
- Overall
  - Ease of use
  - Visual appeal and workflow integration

Data Management / Storage

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Data Management / Storage

- In traditional BI, a special database system called data warehouse or data mart is often used to store enterprise data
- Traditional (operational) relational databases facilitate data management and transaction processing. They have two limitations for data analysis and decision support
  - Performance
  - Heterogeneity
- The data warehouse approach is a centralized and structured approach for analytical data management. For more recent personal BI/analytics, data is also kept locally for easy access and manipulation, without much technical support.

Data Gathering and Integration

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Data Gathering and Integration

- Enterprise level data are coming from multiple different sources, but need to be combined and associated
  - Operational databases
  - Spreadsheets
  - Text, CSV
  - PDF, Paper
- The need to bring together different data/information
  - Autonomous (may not have the control and management of data)
  - Distributed (from different systems and places)
  - Different (in data model, format, or platform)
- General processing steps - ETL
  - Extraction
  - Transformation
  - Loading
- Traditional BI focuses on upfront separate ETL processes that load the data in a centralized storage. In modern BI and analytics, data cleanse and transformation may happen just-in-time with analysis.

Analysis Techniques/Technologies

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- Descriptive reporting
- OLAP (Online Analytical Processing)
- Business analytics
- Advanced and computation intensive: data mining, deep learning, etc.

Basic Techniques in Business Analytics

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- Regression
  - Reasoning, estimating the relationships among variables
- Forecasting
  - Trend analysis, based on extrapolation of historical data
- Correlation
  - Relationship discovery between factors (but not causal relationship)
- Factor analysis
  - Determine impacting variables and their variability

Data Visualization

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Data visualization is the graphical representation and presentation of data for the purpose of perception and understanding

Data visualization in BI

Data visualization is an important part of data exploration and decision making. Given the power of visualization, it is only natural to apply the rich communication techniques in the field of BI and analytics.

Reports

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A report is the presentation of detailed data arranged in defined layouts and formats

Traditional reports contain detailed data in a tabular format and typically display numbers and text only.

Modern reports can be interactive and visual but the focus is still on detailed data. Sometimes the distinction is a bit blurred with dashboards in some practical cases.

Dashboard

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A dashboard is a visual-oriented display of the most important data and information needed to achieve defined goals and objectives; consolidated and arranged on a single screen so the information can be viewed at a glance.

Elements of a dashboard

– Data/information: the most important element

– Visual: data visuals (charts, etc.) provide an high level at-a-glance view – User interface

• a clean UI that unifies all elements to work together as a whole • supporting interactions as needed

Dashboard = data/information + visual + UI

The Modern/New BI

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- A modern BI platform supports IT-enabled analytic content development.
- By contrast, traditional BI platforms are designed to support modular development of IT produced analytic content, and specialized tools and skills and significant upfront data modeling, coupled with a predefined metadata layer, are required to access their analytic capabilities.

The Modern/New BI

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Analytic Workflow Component	Traditional BI Platform	Modern BI Platform
Data source	Upfront dimensional modeling required (IT-built star schemas)	Upfront modeling not required (flat files/flat tables)
Data ingestion and preparation	IT-produced	IT-enabled (business-led)
Content authoring	Primarily IT staff, but also some power users	Business users;
Analysis	Predefined and regular reporting, based on predefined model	Free-form exploration, ad hoc analytics
Insight delivery	Distribution and notifications via scheduled reports or portal; passive collection and use of data (reporting driven).	Sharing and collaboration, storytelling, open APIs

Exercises:

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List at least 3 BI/Analytics Applications, conduct a research, give an example and briefly discuss each.

References

- https://www.researchgate.net/publication/327578485_Business_Intelligence_and_Analytics_A_Comprehensive_Overview
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- – http://pestleanalysis.com/differences-betweenbusiness-analytics-and-business-analysis/

Thank you